Convex combination of two fKFs with weight sharing

2026-10-15

The review combines LMS and NLMS filters. Here its convex combination (cvx-PN-LMS, its eq. (17) and Table 4) and its sharing schemes (Table 5) are written for two fKFs, the filters of Algorithm 2 of the draft. Both methods then run the same two filters, and only the weights and the sharing differ. (Rn) is an equation of the review, (Dn) one of the draft.

1. The component: the fKF as a variable-step NLMS

Filter k has the fixed predicted variance $v^{(k)}$ (D37):

$$e_{t,k} = y_t - \boldsymbol{x}_t^{\mathsf{T}} \boldsymbol{w}_{t-1,k}, \qquad \sigma_{t,k}^2 = v^{(k)} \|\boldsymbol{x}_t\|^2, \qquad \boldsymbol{w}_{t,k} = \boldsymbol{w}_{t-1,k} + \frac{\boldsymbol{x}_t}{\|\boldsymbol{x}_t\|^2}\, \chi_t(e_{t,k}), \tag{1}$$

with χ_t evaluated with $\sigma_{t,k}^2$, by (D43) for Laplacian noise. Written as an NLMS update,

$$\boldsymbol{w}_{t,k} = \boldsymbol{w}_{t-1,k} + \mu_{t,k}\, \frac{\boldsymbol{x}_t\, e_{t,k}}{\|\boldsymbol{x}_t\|^2}, \qquad \mu_{t,k} = \frac{\chi_t(e_{t,k})}{e_{t,k}}. \tag{2}$$

The step $\mu_{t,k}$ changes at every step with the error and with $\sigma_{t,k}$. Since χ_t is odd and $0 \le \chi_t' \le 1$ for log-concave noise, $\mu_{t,k} \in [0, 1]$. For Gaussian noise (D38), $\mu_{t,k} = v^{(k)}\|\boldsymbol{x}_t\|^2/(v^{(k)}\|\boldsymbol{x}_t\|^2 + v_\eta)$, the regularized NLMS. For Laplacian noise $|\chi_t| < \sigma_{t,k}^2/b_\eta$, so an outlier moves $\boldsymbol{w}_{t,k}$ by a bounded amount. The variance $v^{(k)}$ plays the role of the review's step size: filter 1 is the fast one and filter 2 the slow one, $v^{(1)} > v^{(2)}$.

2. The combination: cvx-PN-LMS

The output (R2)–(R4), with the sigmoid (R15):

$$\lambda_t = \mathrm{sgm}(a_t) = \frac{1}{1 + e^{-a_t}}, \qquad \hat{\boldsymbol{w}}_{t-1} = \lambda_t \boldsymbol{w}_{t-1,1} + (1 - \lambda_t)\boldsymbol{w}_{t-1,2},$$
$$e_t = y_t - \boldsymbol{x}_t^{\mathsf{T}} \hat{\boldsymbol{w}}_{t-1} = \lambda_t e_{t,1} + (1 - \lambda_t) e_{t,2}. \tag{3}$$

With $\hat{y}_{t,k} = \boldsymbol{x}_t^{\mathsf{T}} \boldsymbol{w}_{t-1,k}$, $\partial e_t/\partial \lambda_t = -(\hat{y}_{t,1} - \hat{y}_{t,2})$ and $d\lambda_t/da_t = \lambda_t(1 - \lambda_t)$. A stochastic gradient step on e_t^2, normalized by the power of $\hat{y}_{t,1} - \hat{y}_{t,2}$, gives (R17), Table 4:

$$p_t = \gamma\, p_{t-1} + (1 - \gamma)(\hat{y}_{t,1} - \hat{y}_{t,2})^2, \tag{4}$$

$$a_{t+1} = \Big[\, a_t + \frac{\mu_a}{p_t}\, \lambda_t(1 - \lambda_t)\, e_t\, (\hat{y}_{t,1} - \hat{y}_{t,2}) \Big]_{-a^+}^{a^+}, \tag{5}$$

where $[\,\cdot\,]_{-a^+}^{a^+}$ clips to $[-a^+, a^+]$ ($a^+ = 4$ in the review). γ is the review's η, renamed because η_t is the noise here. $p_t = 0$ only while $\hat{y}_{t,1} = \hat{y}_{t,2}$, for example while both filters are still at $\mathbf{0}$; a_t is not moved then. Nothing in (3)–(5) depends on the component filters. Replacing the review's LMS filters by fKFs changes only how $\hat{y}_{t,k}$ and $\boldsymbol{w}_{t,k}$ are produced.

3. Weight sharing: the review's Table 5

After both filters' step (1), with this step's λ_t:

scheme	condition	rule	parameters
none			
gradual transfer	$\lambda_t \geq \lambda_0$	$\boldsymbol{w}_{t,2} \leftarrow \ell\, \boldsymbol{w}_{t,2} + (1-\ell)\, \boldsymbol{w}_{t,1}$	$0 < \ell < 1$, $0 \ll \lambda_0 < 1$
simple copy	$\lambda_t \geq \lambda_0$ and $t = mN_0$	$\boldsymbol{w}_{t,2} \leftarrow \boldsymbol{w}_{t,1}$	$N_0 \geq 2$, $0 \ll \lambda_0 < 1$
feedback	$t = mN_0$	$\boldsymbol{w}_{t,1}, \boldsymbol{w}_{t,2} \leftarrow \lambda_t \boldsymbol{w}_{t,1} + (1-\lambda_t)\boldsymbol{w}_{t,2}$	$N_0 \geq 2$

The fKF carries no state other than $\boldsymbol{w}_{t,k}$, so sharing $\boldsymbol{w}$ shares the whole filter. With the sKFs of `cvx-combination-vs-bank.ipynb` each filter kept its own v. The schemes are written from the review only; its references [47]–[49] were not read.

4. Against Algorithm 2 with $K = 2$

Both methods run the two fKF steps (1). They differ in two places.

The weight. The combination learns λ_t by the gradient (5) on the squared output error. The bank uses the posterior $\mu_{t,j} \propto c_{t,j}\, f_{t,j}(e_{t,j})$ (D56), from each filter's predictive density at its own error.

The sharing. The bank shares through its mixing (D53)–(D54), at every step and before the update. With $K = 2$ the transitions (D52) are $p_{12} = p_{21} = q$, and the mixing for filter 2 reads

$$\bar{\boldsymbol{w}}_{t,2} = \ell_{t,2}\, \boldsymbol{w}_{t-1,2} + (1-\ell_{t,2})\, \boldsymbol{w}_{t-1,1}, \qquad \ell_{t,2} = \frac{(1-q)\,\mu_{t-1,2}}{(1-q)\,\mu_{t-1,2} + q\,\mu_{t-1,1}}, \tag{6}$$

and the same with the indices swapped for filter 1. This is the gradual transfer of Table 5 with three differences: $\ell_{t,j}$ is set at every step by the weights and q, not fixed by hand; there is no threshold, since $\ell_{t,2}$ is close to 1 while the slow filter is probable and drops when the fast one takes over; and it runs in both directions. Each filter then steps from $\bar{\boldsymbol{w}}_{t,j}$, not from $\boldsymbol{w}_{t-1,j}$.

Algorithm 1 Convex combination of two fKF filters (cvx-PN-LMS) with weight sharing

Require: fixed predicted variances $v^{(1)} > v^{(2)}$ (fast, slow), step μ_a, forgetting factor γ, bound a^+, noise scale b_η; a sharing scheme and its parameters (ℓ, λ_0, N_0)

$\boldsymbol{w}_{0,k} \leftarrow \mathbf{0}$ for $k = 1, 2$; $a_1 \leftarrow 0$, $\lambda_1 \leftarrow 1/2$, $p_0 \leftarrow 0$
for $t = 1, 2, \ldots$ **do**
  receive $\boldsymbol{x}_t$ and y_t
  for $k = 1, 2$ **do** ▷ one fKF step per filter
    $\sigma^2 \leftarrow v^{(k)} \|\boldsymbol{x}_t\|^2$, $\hat{y}_k \leftarrow \boldsymbol{x}_t^{\mathsf{T}} \boldsymbol{w}_{t-1,k}$, $e_k \leftarrow y_t - \hat{y}_k$
    $\boldsymbol{w}_{t,k} \leftarrow \boldsymbol{w}_{t-1,k} + \boldsymbol{x}_t \, \chi_t(e_k) / \|\boldsymbol{x}_t\|^2$
  end for
  $e \leftarrow y_t - \lambda_t \hat{y}_1 - (1 - \lambda_t)\hat{y}_2$, $p_t \leftarrow \gamma p_{t-1} + (1 - \gamma)(\hat{y}_1 - \hat{y}_2)^2$ ▷ cvx-PN-LMS, (3)–(4)
  $a_{t+1} \leftarrow a_t + \mu_a \lambda_t (1 - \lambda_t)\, e\, (\hat{y}_1 - \hat{y}_2)/p_t$ if $p_t > 0$, else a_t; clip to $[-a^+, a^+]$ ▷ (5)
  if gradual transfer and $\lambda_t \geq \lambda_0$ **then** ▷ sharing, Table 5
    $\boldsymbol{w}_{t,2} \leftarrow \ell \, \boldsymbol{w}_{t,2} + (1 - \ell) \, \boldsymbol{w}_{t,1}$
  else if simple copy and $\lambda_t \geq \lambda_0$ and $t = mN_0$ **then**
    $\boldsymbol{w}_{t,2} \leftarrow \boldsymbol{w}_{t,1}$
  else if feedback and $t = mN_0$ **then**
    $\boldsymbol{w}_{t,1}, \boldsymbol{w}_{t,2} \leftarrow \lambda_t \boldsymbol{w}_{t,1} + (1 - \lambda_t)\boldsymbol{w}_{t,2}$
  end if
  $\lambda_{t+1} \leftarrow \mathrm{sgm}(a_{t+1})$, $\hat{\boldsymbol{w}}_t \leftarrow \lambda_{t+1} \boldsymbol{w}_{t,1} + (1 - \lambda_{t+1}) \boldsymbol{w}_{t,2}$ ▷ output
end for

The output $\hat{\boldsymbol{w}}_t$ is formed after the sharing, with the new weight λ_{t+1}, as $\hat{\boldsymbol{w}}_t$ of Algorithm 2 is formed with $\mu_{t,j}$. It is the vector that predicts y_{t+1}. The algorithm takes the operations of two fKFs plus $O(M)$ for the combination, and $O(M)$ more on the steps where a scheme shares.